Helicons

EM wave in electron gas subject to external magnetic field B_0

$$\varepsilon_{ij}(\omega) = \varepsilon_0 \delta_{ij} + \frac{i}{\omega}\sigma_{ij}(\omega)$$

We found
$$\sigma^{\perp}_{ij}(\omega) = \frac{e^2 n}{m}\begin{pmatrix} -i\omega + \frac{1}{\tau} & -\Omega_c \\ \Omega_c & -i\omega + \frac{1}{\tau}\end{pmatrix}\frac{1}{\Omega_c^2 + (-i\omega + \frac{1}{\tau})^2}$$

$$\sigma_{zz}(\omega) = \frac{e^2 n}{m}\frac{1}{-i\omega + \frac{1}{\tau}} \quad ; \text{ also } \omega_p^2 = \frac{e^2 n}{\varepsilon_0 m}$$

$$\Rightarrow \frac{i\sigma_{zz}}{\omega} = \varepsilon_0 \frac{i\omega_p^2}{\omega(-i\omega + \frac{1}{\tau})}$$

$$\frac{i\sigma^{\perp}}{\omega} = \varepsilon_0 \omega_p^2 \begin{pmatrix} 1 + \frac{i}{\omega\tau} & -\frac{i\Omega_c}{\omega} \\ \frac{i\Omega_c}{\omega} & 1 + \frac{i}{\omega\tau}\end{pmatrix}\frac{1}{\Omega_c^2 + (-i\omega + \frac{1}{\tau})^2}$$

We take the limit $\frac{1}{\tau} \ll \omega \ll \Omega_c \ll \omega_p$

Then
$$\frac{1}{\Omega_c^2 + (-i\omega + \frac{1}{\tau})^2} \approx \frac{1}{\Omega_c^2 - \omega^2} \approx \frac{1}{\Omega_c^2}$$

$$\frac{1}{\omega\tau} \ll 1.$$

$$\Rightarrow \varepsilon^{\perp}(\omega) \cong \cancel{\varepsilon_0 \delta_{ij}} + \varepsilon_0 \begin{pmatrix} \frac{\omega_p^2}{\Omega_c^2} & \frac{-i\omega_p^2}{\Omega_c \omega} \\ \frac{i\omega_p^2}{\Omega_c \omega} & \frac{\omega_p^2}{\Omega_c^2}\end{pmatrix}$$

and we can neglect $\varepsilon_0 \delta_{ij}$ in comparison with $\left(\frac{\omega_p}{\Omega_c}\right)^2 \gg 1$.

Also, $\varepsilon^{\parallel}(\omega) = \varepsilon_{zz}(\omega) = \varepsilon_0 \delta_{ij} + \varepsilon_0 \frac{i\omega_p^2}{\omega(-i\omega + \frac{1}{\tau})} \approx -\varepsilon_0 \frac{\omega_p^2}{\omega^2}$

Observe that $\frac{\omega_p^2}{\Omega_c \omega} = \left(\frac{\omega_p}{\Omega_c}\right)^2 \frac{\Omega_c}{\omega} \gg \left(\frac{\omega_p}{\Omega_c}\right)^2$

$$\Rightarrow \varepsilon_{xx} \ll \varepsilon_{xy} \ll \varepsilon_{zz}$$

$$\begin{cases} \vec{k}\times\vec{B} = -\mu_0\omega\vec{D} = -\dfrac{\omega}{\varepsilon_0 c^2}\vec{D} \\ \vec{k}\times\vec{E} = \omega\vec{B} \end{cases}$$

$$\Rightarrow \underbrace{\vec{k}\times(\vec{k}\times\vec{E})}_{\vec{k}(\vec{k}\cdot\vec{E})-\vec{E}k^2} = -\frac{\omega^2}{\varepsilon_0 c^2}\vec{D}$$

$$\Rightarrow \left(k^2\delta_{ij} - k_i k_j\right)E_j = \frac{\omega^2}{\varepsilon_0 c^2}\underbrace{\varepsilon_{ij}(\omega)E_j}_{D_i}$$

Let us consider wave propagating along $\vec{B}_0 \Rightarrow \vec{k} = (0,0,k)$
$\hookrightarrow B_0\hat{z}$.

$$\Rightarrow (k^2 - k_z^2)E_z = 0 \Rightarrow D_z = 0 \Rightarrow E_z = 0.$$

The wave is transverse $\vec{E} = (E_x, E_y, 0)$, $\vec{k}\cdot\vec{E} = 0$.

$$k^2 E_x = \frac{\omega^2}{\varepsilon_0 c^2}\left(\varepsilon_{xx}E_x + \varepsilon_{xy}E_y\right) =$$

$$= \frac{\omega^2}{\varepsilon_0 c^2}\varepsilon_0\left(\frac{\omega_p^2}{\Omega_c^2}E_x + \frac{-i\omega_p^2}{\Omega_c\omega}E_y\right)$$

$$\begin{cases} \left(k^2 - \dfrac{\omega^2\omega_p^2}{c^2\Omega_c^2}\right)E_x + \dfrac{i\omega\omega_p^2}{c^2\Omega_c}E_y = 0 \\ \left(k^2 - \dfrac{\omega^2\omega_p^2}{c^2\Omega_c^2}\right)E_y - \dfrac{i\omega\omega_p^2}{c^2\Omega_c}E_x = 0 \end{cases}$$

$$\begin{bmatrix} & \\ & \end{bmatrix}\begin{pmatrix} E_x \\ E_y \end{pmatrix} = 0 \Rightarrow \det\begin{bmatrix} & \\ & \end{bmatrix} = 0$$

$$\Rightarrow \left(k^2 - \frac{\omega^2 \omega_p^2}{c^2 \Omega_c^2}\right)^2 - \left(\frac{\omega \omega_p^2}{c^2 \Omega_c}\right)^2 = 0$$

$$k^2 - \frac{\omega^2 \omega_p^2}{c^2 \Omega_c^2} = \pm \frac{\omega \omega_p^2}{c^2 |\Omega_c|}$$

But $\frac{\omega_p^2}{c^2}\left(\frac{\omega}{\Omega_c}\right)^2 \ll \frac{\omega_p^2}{c^2}\frac{\omega}{\Omega_c}$

$\Rightarrow$ can neglect $\left(\frac{\omega \omega_p}{c \Omega_c}\right)^2$!

$$\Rightarrow \omega = c^2 k^2 \frac{\Omega_c}{\omega_p^2}$$ (only $\oplus$ sign is physical for $\Omega_c > 0$)

1) The obtained dispersion is quadratic, $\omega \sim k^2$.
And the frequency is quite low since $\frac{\Omega_c}{\omega_p} \ll 1$.

2)
$$\begin{pmatrix} k^2 & \frac{i\omega \omega_p^2}{c^2 \Omega_c} \\ -\frac{i\omega \omega_p^2}{c^2 \Omega_c} & k^2 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = 0$$

$$\Rightarrow E_x = -\frac{i\omega \omega_p^2}{c^2 k^2 \Omega_c} E_y = -i E_y \quad \text{for } \Omega_c > 0$$

(and $E_x = i E_y$ for $\Omega_c < 0$)

$\Rightarrow$ the propagating EM wave is circularly polarized!

Right-handed polarization for $\Omega_c > 0$ (electrons)

$\rightarrow$ helicon (in plasma physics $\rightarrow$ whistling atmospherics)

Can be thought of as an "extreme" version of Faraday rotation.

Recall The usual (no magnetic field, $B_0 = 0$) limit:

$$\Omega_c = 0 \Rightarrow \sigma_{ij} = \frac{e^2 n}{m} \begin{pmatrix} -i\omega + \frac{1}{\tau} & 0 \\ 0 & -i\omega + \frac{1}{\tau} \end{pmatrix} \frac{1}{\left(-i\omega + \frac{1}{\tau}\right)^2} = \frac{e^2 n \tau}{m} \frac{1}{1 - i\omega\tau} \delta_{ij}$$

is diagonal!

$$\Rightarrow \varepsilon_{ij}(\omega) = \varepsilon_0 \delta_{ij} + \frac{i\sigma_{ij}(\omega)}{\omega} = \left(\varepsilon_0 + \frac{i\sigma_0}{\omega(1 - i\omega\tau)}\right)\delta_{ij}$$

$$\frac{\varepsilon(\omega)}{\varepsilon_0} = 1 + \frac{i\sigma_0}{\varepsilon_0 \omega} \frac{1}{-i\omega\tau\left(1 + \frac{i}{\omega\tau}\right)} = 1 - \frac{\sigma_0}{\varepsilon_0 \tau \omega^2} \frac{1}{1 + \frac{i}{\omega\tau}}$$

$\Rightarrow$ For $\omega\tau \gg 1$ $\quad \frac{\varepsilon(\omega)}{\varepsilon_0} \simeq 1 - \frac{e^2 n}{\varepsilon_0 m} \frac{1}{\omega^2} = 1 - \frac{\omega_p^2}{\omega^2}$

so that $n^2(\omega) = \frac{\varepsilon(\omega)}{\varepsilon_0} = -\left(\frac{\omega_p^2}{\omega^2} - 1\right) < 0$ for $\omega < \omega_p$

$\Rightarrow$ the wave eqn $\quad k^2 \vec{E} = \frac{\omega^2}{\varepsilon_0 c^2} \varepsilon(\omega) \vec{E} = \frac{\omega^2 n^2(\omega)}{c^2} \vec{E}$

$$\Rightarrow k^2 = -\frac{\omega^2}{c^2}\left(\frac{\omega_p^2}{\omega^2} - 1\right) = \frac{-1}{c^2}\left(\omega_p^2 - \omega^2\right)$$

So that $k = i\varkappa$ is purely imaginary $\Rightarrow \varkappa = \frac{1}{c}\sqrt{\omega_p^2 - \omega^2}$ and describes transverse wave which decay exponentially fast into the bulk.

The helicon $\left(\Omega_c = \frac{eB_0}{m} > \omega,\ \Omega_c \ll \omega_p\right)$ describes the same medium subject to strong magnetic field B_0 and now the EM wave can propagate through the conductor!

Hall Effect

The Hall current: $\vec{j}$ perpendicular to $\vec{B}_0$ and $\vec{E}_\perp$

<u>Static limit</u> $\omega = 0$

$$\sigma^\perp(0) = \frac{e^2 n}{m} \frac{1}{\Omega_c^2 + \frac{1}{\tau^2}} \begin{pmatrix} \frac{1}{\tau} & -\Omega_c \\ \Omega_c & \frac{1}{\tau} \end{pmatrix} =$$

$$= \underbrace{\frac{e^2 n \tau}{m}}_{\sigma_0} \frac{1}{1+(\Omega_c \tau)^2} \begin{pmatrix} 1 & -\Omega_c \tau \\ \Omega_c \tau & 1 \end{pmatrix}$$

Hall conductivity $\sigma_H = \sigma_{xy} = \sigma_{12} = \sigma_0 \dfrac{-\Omega_c \tau}{1+(\Omega_c \tau)^2}$

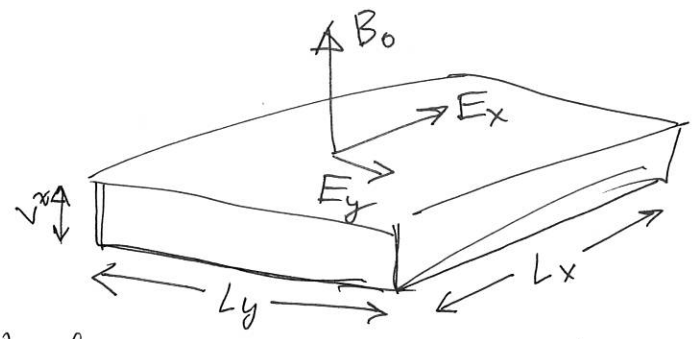
$$\xrightarrow[\Omega_c \tau \gg 1]{} -\frac{\sigma_0}{\Omega_c \tau} = -\frac{e^2 n \tau}{m} \frac{m}{e B_0 \tau} =$$

$$= -\frac{en}{B_0}$$

is independent of m, τ.

Depends on the sign of charge carriers: $(-e)$ for electrons,
$(+e)$ for holes.

Hall effect



Apply B_0 and E_x $\Rightarrow$ then current flows along $\hat{x}$ and $\hat{y}$ $\Rightarrow$ but current along $\hat{y}$ can not flow indefenitely as charges accumulate on the ~~sides~~ sides of the sample and result in E_y which stops the flow of j_y

$$\Rightarrow \quad j_y = 0 \quad , \text{but} \quad j_y = \sigma_{yx} E_x + \sigma_{yy} E_y$$

$$\Rightarrow \quad E_y = -\frac{\sigma_{yx}}{\sigma_{yy}} E_x$$

$$j_x = \sigma_{xx} E_x + \sigma_{xy} E_y = \left(\sigma_{xx} - \frac{\sigma_{xy}\sigma_{yx}}{\sigma_{yy}}\right) E_x$$

$$= \frac{E_x}{\rho_{xx}}$$

Resistivity tensor $\rho = \sigma^{-1}$

$$\begin{pmatrix} \rho_{xx} & \rho_{xy} \\ \rho_{yx} & \rho_{yy} \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix}^{-1} = \frac{1}{\sigma_{xx}\sigma_{yy} - \sigma_{xy}\sigma_{yx}} \begin{pmatrix} \sigma_{yy} & -\sigma_{xy} \\ -\sigma_{yx} & \sigma_{xx} \end{pmatrix}$$

$$\Rightarrow \rho_{xx} = \frac{\sigma_{yy}}{\sigma_{xx}\sigma_{yy} - \sigma_{xy}\sigma_{yx}} = \frac{1}{\sigma_{xx} - \frac{\sigma_{xy}\sigma_{yx}}{\sigma_{yy}}}$$

$$\sigma(0) = \frac{\sigma_0}{1+(\Omega_c\tau)^2}\begin{pmatrix} 1 & -\Omega_c\tau \\ \Omega_c\tau & 1 \end{pmatrix} \quad , \quad \sigma_{xy} = -\sigma_{yx}$$

$$\sigma_{xx} = \sigma_{yy}$$

Hence $\sigma_{xx} - \dfrac{\sigma_{xy}\sigma_{yx}}{\sigma_{yy}} = \sigma_{xx} + \dfrac{\sigma_{xy}^2}{\sigma_{xx}} = \dfrac{\sigma_0}{1+(\Omega_c\tau)^2} + \dfrac{\left(\dfrac{\sigma_0\Omega_c\tau}{1+(\Omega_c\tau)^2}\right)^2}{\dfrac{\sigma_0}{1+(\Omega_c\tau)^2}}$

$$= \frac{\sigma_0}{1+(\Omega_c\tau)^2} + \frac{\sigma_0(\Omega_c\tau)^2}{1+(\Omega_c\tau)^2} = \sigma_0$$

$$\Rightarrow \rho_{xx} = \rho_{yy} = \frac{1}{\sigma_0}$$

$$\rho = \rho_0\begin{pmatrix} 1 & (\Omega_c\tau)^{-1} \\ -(\Omega_c\tau)^{-1} & 1 \end{pmatrix} \quad , \rho_0 = \frac{1}{\sigma_0}$$

$\Rightarrow j_x = \sigma_0 E_x$ is not effected by B_0 !

Hall effect: $R_H = \dfrac{V_y}{I_x} = \dfrac{E_y L_y}{j_x L_y L_z} = \dfrac{E_y}{j_x L_z} = -\dfrac{E_x \sigma_{yx}}{j_x \sigma_{yy} L_z}$

$$= -\frac{E_x \sigma_{yx}}{\sigma_0 E_x \sigma_{yy} L_z} = -\frac{\sigma_{yx}}{\sigma_0 \sigma_{yy} L_z} = \frac{-\Omega_c\tau}{\sigma_0 L_z} =$$

$$= \frac{-\frac{eB_0}{m}\tau}{\frac{e^2 n\tau}{m} L_z} = -\frac{B_0}{enL_z}$$